


Problem 1

Simple example. $|Z| = 3$, $Z = \{a, b, c\}$ $a < b < c$ ($v(a) < v(b) < v(c)$)

$$p = \frac{1}{2} [c] \oplus \frac{1}{2} [a] \Rightarrow U(p) = \alpha v(c) + (1-\alpha) v(a)$$

$$q = \frac{1}{3} [c] \oplus \frac{1}{3} [b] + \frac{1}{3} [a] \Rightarrow U(q) = \alpha v(c) + (1-\alpha) v(a)$$

Continuity. No. (Same setup as above)

Consider $p_n = \begin{pmatrix} 1/n & 1/n & 1-2/n \\ a & b & c \end{pmatrix}$ for each $n=1, 2, \dots$

$$p = (0, 0, 1) \quad p_n \xrightarrow{n \rightarrow \infty} p$$

$$U(p_n) = \alpha v(c) + (1-\alpha) v(a) \text{ for every } n$$

$$U(p) = \alpha v(c) + (1-\alpha) v(c) = v(c) \quad v(c) \neq \alpha v(c) + (1-\alpha) v(a)$$

$\therefore$ Not Continuous

Independence NO. Same setup as above

Consider $L = \frac{1}{2} [c] \oplus \frac{1}{2} [b]$

$$L' = \frac{1}{2} [b] \oplus \frac{1}{2} [a]$$

$$U(L) = \alpha v(c) + (1-\alpha)v(b) \Rightarrow U(L) > U(L') \Leftrightarrow L \succ L'$$

$$U(L') = \alpha v(b) + (1-\alpha)v(a)$$

Consider $L'' = \frac{1}{2} [c] + \frac{1}{2} [a]$

$$\frac{1}{2} L \oplus \frac{1}{2} L'' = \frac{1}{2} [c] \oplus \frac{1}{4} [b] \oplus \frac{1}{4} [a] \Rightarrow U(\frac{1}{2} L \oplus \frac{1}{2} L'') = \alpha v(c) + (1-\alpha)v(a)$$

$$\frac{1}{2} L' \oplus \frac{1}{2} L'' = \frac{1}{4} [c] \oplus \frac{1}{4} [b] \oplus \frac{1}{2} [a] \Rightarrow U(\frac{1}{2} L' \oplus \frac{1}{2} L'') = \alpha v(c) + (1-\alpha)v(a)$$

$$\Rightarrow \frac{1}{2} L \oplus \frac{1}{2} L'' \sim \frac{1}{2} L' \oplus \frac{1}{2} L'' \quad \text{which violates}$$

independence

$\therefore U$ doesn't admit an EU representation.

(b) p a lottery, E_p : expected value of p
 $U([E_p])$ vs $U(p)$

Let a, c be such that $a < c$. For any α we can find, by the continuity of u , a number $b \in (a, c)$ such that
$$\alpha u(c) + (1-\alpha)u(a) > u(b).$$

Now, let λ be the number such that $\lambda c + (1-\lambda)a < b$

Consider the lottery $p = \lambda [c] \oplus (1-\lambda) [a]$. Note $E_p = \lambda c + (1-\lambda)a$

$$U([E_p]) = \alpha u(E_p) + (1-\alpha)u(E_p) = u(E_p) = u(\lambda c + (1-\lambda)a) < u(b) \quad \text{since } u \text{ is increasing}$$

$$\text{but } u(b) < \alpha u(c) + (1-\alpha)u(a) = U(p)$$

$\therefore U([E_p]) < U(p) \Rightarrow$ since this construction is valid for $\alpha \in (0, 1]$, for these α , U doesn't exhibit risk aversion.

If $\alpha=0$,
$$U(p) = \min \{v(z) \mid p(z) > 0\}$$

for any $p \in \mathcal{L}(Z)$
$$U(p) \leq U([E_p])$$

$$\begin{array}{ccc} \uparrow & & \nwarrow \\ \min \{v(z) \mid p(z) > 0\} & & v(E_p) \end{array}$$

For the minimal $\hat{z}$
with $p(\hat{z}) > 0$

$\hat{z} < E_p$

$\Rightarrow v(\hat{z}) < v(E_p)$

U exhibits risk aversion

Problem 2

p - price of y (random)

w_1, w_2 - price of inputs

$$TC = y^2(w_1 + w_2)$$

Firm's problem $\max_{y \geq 0} E[p \cdot y - y^2(w_1 + w_2)] = \max_{y \geq 0} (E[p] \cdot y - y^2(w_1 + w_2))$

$$F.O.C.: E(p) - 2y^*(w_1 + w_2) = 0 \Rightarrow y^* = \frac{E(p)}{2(w_1 + w_2)}$$

$$S.O.C.: -2(w_1 + w_2) < 0 \quad \nearrow \text{the maximum.}$$

$$(a) \quad E(p) = \lambda p_h + (1-\lambda)p_l$$

$$(b) \quad E(p) = \int_{p_l}^{p_h} p f(p) dp$$

Problem 3

(a) $N=2$

Prob. (pmf)	Residual Loss	
$(1-p)^2$	Nobody loses	0
$2p(1-p)$	1 Loses	$50 = \frac{100}{2}$
p^2	2 Loses	100

$$EV = 2p(1-p) \cdot 50 + p^2 \cdot 100$$

$$= 100 [p(1-p) + p^2]$$

$$= 100p$$

$N=3$

prob. (pmf)	Residual Loss	
$(1-p)^3$	Nobody loses	0
$3p(1-p)^2$	1 Lose, 2 doesn't	$\frac{100}{3} \approx 33.33$
$3p^2(1-p)$	2 Lose, 1 doesn't	$\frac{200}{3} \approx 66.66$
p^3	3 Lose	100

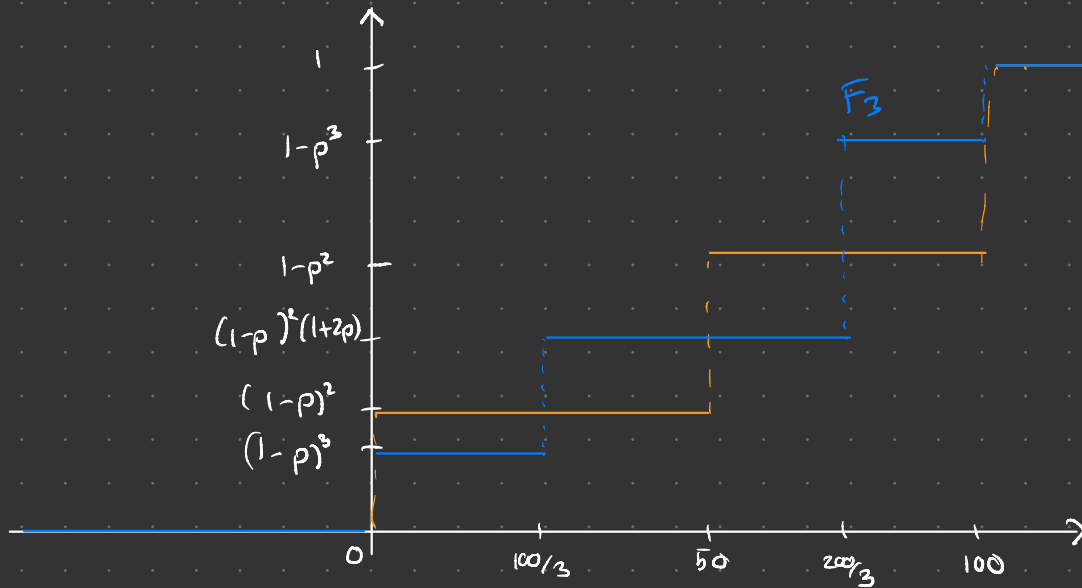
$$EV = 3p(1-p)^2 \frac{100}{3} + 3p^2(1-p) \frac{200}{3} + p^3 100$$

$$= 100p$$

(b)

$$F_2(x) = \begin{cases} 0 & x < 0 \\ (1-p)^2 & 0 \leq x < 50 \\ 1-p^2 & 50 \leq x < 100 \\ 1 & x \geq 100 \end{cases}$$

$$F_3(x) = \begin{cases} 0 & x < 0 \\ (1-p)^3 & 0 \leq x < 33.33 \\ (1-p)^2(1+2p) & 33.33 \leq x < 66.66 \\ 1-p^3 & 66.66 \leq x < 100 \\ 1 & x \geq 100 \end{cases}$$



$$\int_{-\infty}^t F_2(x) dx \geq \int_{-\infty}^t F_3(x) dx$$

For every t

R-S condition.

Is immediate that $\int_{-\infty}^{100} F_3(x) dx = \int_{-\infty}^{100} F_2(x) dx$ because they have the same mean //

Let's generalize a bit.

Define a R.V. $X_i = \begin{cases} 1 & \text{with } p \\ 0 & \text{with } (1-p) \end{cases}$ $X_i = \begin{matrix} \text{R.V.} \\ \text{Lose} \\ \text{or} \\ \text{not} \end{matrix}$ each X_i is independent of X_j
 $i = \text{agent}$

N agents $X(N) = X_1 + X_2 + \dots + X_N$ (I changed a little the notation with respect to the video)

$X(N)$ is the sum of N Bernoulli(p) R.V.'s $X(N) \sim \text{Bin}(p, N)$

$Z_N = \frac{100}{N} X(N)$ R.V. that represents the residual loss

$$\mathbb{E}(Z_N) = \frac{100}{N} \mathbb{E}(X(N)) = \frac{100}{N} pN = 100p //$$

I will show that

Z_N is a mean preserving spread of Z_{N+1}
(X_N is a mean-preserving spread of X_{N+1})

$$\begin{aligned} Z_{N+1} &= \frac{100}{N+1} X_{(N+1)} = \frac{100}{N+1} \left[\sum_{i=1}^{N+1} X_i \right] = \frac{100}{N+1} \left[\sum_{i=1}^N X_i + X_{N+1} \right] \\ &= \frac{100}{N+1} [X(N) + X_{N+1}] = \frac{100}{N+1} X(N) \cdot \frac{N}{N} + \frac{100}{N+1} X_{N+1} \\ &= Z_N \cdot \frac{N}{N+1} + \frac{100}{N+1} X_{N+1} \end{aligned}$$

$$\text{Thus } Z_{N+1} = Z_N \left(\frac{N}{N+1} \right) + \frac{100}{N+1} X_{N+1}$$

We can write:

$$\left(\frac{N+1}{N} \right) Z_{N+1} = Z_N + \frac{100}{N} X_{N+1} \Leftrightarrow$$

$$Z_{N+1} + \frac{Z_{N+1}}{N} - \frac{100}{N} X_{N+1} = Z_N$$

$$\text{Thus: } Z_N = Z_{N+1} + \frac{1}{N} (Z_{N+1} - 100 X_{N+1})$$

Let $\varepsilon = \frac{1}{N} (Z_{N+1} - 100 X_{N+1})$. This shows that Z_N can be decomposed as

$$Z_N = Z_{N+1} + \varepsilon$$

It is then enough to show that

$$\mathbb{E}(\varepsilon | Z_{N+1}) = 0.$$

Let's do this:

$$\mathbb{E} \left[\frac{1}{N} (Z_{N+1} - 100 X_{N+1}) \mid Z_{N+1} \right] =$$

$$\frac{1}{N} \mathbb{E} \left[\frac{100 X(N+1)}{N+1} - 100 X_{N+1} \mid Z_{N+1} \right] =$$

$$\frac{100}{N} \mathbb{E} \left[\frac{\sum_{i=1}^{N+1} X_i}{N+1} - X_{N+1} \mid Z_{N+1} \right] =$$

$$\frac{100}{N} \mathbb{E} \left[\frac{\sum_{i=1}^{N+1} X_i - (N+1) X_{N+1}}{N+1} \mid Z_{N+1} \right] =$$

$$= \frac{100}{N} \mathbb{E} \left[\frac{\sum_{i=1}^{N+1} (X_i - X_{N+1})}{N+1} \mid Z_{N+1} \right] = \frac{100}{N(N+1)} \mathbb{E} \left[\sum_{i=1}^{N+1} (X_i - X_{N+1}) \mid Z_{N+1} \right]$$

$$= \frac{100}{N(N+1)} \sum_{i=1}^{N+1} \mathbb{E} [X_i - X_{N+1} \mid Z_{N+1}]$$

Thus it is enough to show that, for each $i=1, 2, \dots, N+1$, we have that

$\mathbb{E} [X_i - X_{N+1} \mid Z_{N+1}] = 0$. This is equivalent to showing that

$\mathbb{E} [X_i \mid Z_{N+1}] = \mathbb{E} [X_{N+1} \mid Z_{N+1}]$. This last statement is true because the

Random Variables $X_i, i=1, \dots, N+1$ are all independent and identically distributed.

With this last result we have that $\mathbb{E}(\varepsilon \mid Z_{N+1}) = 0$ and thus

Z_N is a mean-preserving spread of Z_{N+1} .

$$\therefore Z_{N+1} \triangleright_{\text{msp}} Z_N.$$

